

MATERIAL SAFETY DATA SHEET

SECTION 1. CHEMICAL IDENTIFICATION

Product Name:	Fipronil 80% WG
Company Name:	Ningbo Sunjoy Agrosience Co., Ltd.
Common Name of A.I.:	Fipronil
Chemical Name:	(±)-5-amino-1-(2,6-dichloro- α,α,α -trifluoro- <i>p</i> -tolyl) -4-trifluoromethylsulfinylpyrazole-3-carbonitrile (IUPAC)
Synonym:	None
Chemical Family:	Phenylpyrazole
Chemical Formula:	C ₁₂ H ₄ Cl ₂ F ₆ N ₄ OS
Use:	Insecticide
Formulation:	WG (Water dispersible granule)

SECTION 2. COMPOSITION/INFORMATION ON INGREDIENTS

Component	Cas No.	Concentration (w/w)
Fipronil	120068-37-3	80% Min
Other inert ingredients	/	20% Max

Ingredients not precisely identified are proprietary or non-hazardous.

SECTION 3. HAZARDS IDENTIFICATION

Hazard Classification:

Hazardous. Toxic, dangerous for the environment.

Risk Phrases:

R23/25 Toxic by inhalation and if swallowed.

R48/25 Toxic: danger of serious damage to health by prolonged exposure if swallowed.

R50/53 Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

Safety Phrases:

S1/2 Keep locked up and out of reach of children.

S13 Keep away from food, drink and animal feeding stuffs.

S20/21 When using, do not eat, drink or smoke.

S35 This material and its container must be disposed of in a safe way.

S46 If swallowed, seek medical advice immediately and show this container or label.

S57 Use appropriate container to avoid environmental contamination.

SECTION 4. FIRST AID MEASURES

First Aid Measures

Ingestion: If swallowed: Call a poison control center or doctor immediately for treatment advice. Have the person sip a glass of water if able to swallow. Do not induce vomiting unless told to do so by a poison control center or doctor.

Eye Contact: If in eyes: Hold eye open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after 5 minutes, then continue rinsing eye. Call a poison control center or doctor for treatment advice.

Skin Contact: If on skin or clothing: Take off contaminated clothing. Rinse skin immediately with plenty of water for 15-20 minutes. Call a poison control center or doctor for treatment advice.

Inhalation: If inhaled: Move person to fresh air. If person is not breathing, call an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call a poison control center or doctor for further treatment advice.

Note to Physician

Treat symptomatically. There is no specific antidote.

SECTION 5. FIRE FIGHTING MEASURES
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Fire and Explosion

Flash Point: Not available

Autoignition Temperature: Not available

Flammability: Not flammable

Unusual Fire, Explosion and Reactivity Hazards

During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion.

In Case of Fire

Use dry chemical, foam or CO₂ extinguishing media. Wear full protective clothing and self-contained breathing apparatus. Evacuate nonessential personnel from the area to prevent human exposure to fire, smoke, fumes or products of combustion. Prevent use of contaminated buildings, area, and equipment until decontaminated. Water runoff can cause environmental damage. If water is used to fight fire, dike and collect runoff.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Emergency Procedures

In case of spillage it is important to take all steps necessary to:

- Avoid eye and skin contact.
- Avoid contamination of waterways

Methods and Materials for Containment and Clean Up

Procedure for spill:

- (1) Keep all bystanders away.
- (2) Wear goggles, half face-piece respirator with combined dust and vapour cartridge, full length clothing and PVC gloves.
- (3) Reposition any leaking containers so as to minimise further leakage.
- (4) Dam and absorb spill with an absorbent material (e.g. sand or soil).
- (5) Shovel the absorbed spill into drums.
- (6) Disposal of the absorbed material will depend upon the extent of the spill.
 - For quantities up to 50kg of product bury in a secure landfill site.
 - For quantities greater than 50kg seek advice (use emergency contact number below) from the manufacturer before attempting disposal. Contain in a secure location until disposal method is established.
- (7) Decontaminate the spill area with detergent and water and rinse with water

SECTION 7. HANDLING AND STORAGE

Store the material in a well-ventilated, secure area out of reach of children and domestic animals. Do not store food, beverages or tobacco products in the storage area. Prevent eating, drinking, tobacco use, and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling.

SECTION 8. EXPOSURE CONTROLS/ PERSONAL PROTECTION

Exposure Standards

Not set.

Engineering Controls

If airborne mists or vapors are generated, use local exhaust ventilation controls.

Personal Protection

Harmful if swallowed. Will irritate the eyes and skin. Avoid contact with eyes and skin. If product in eyes, wash it out immediately with water. When opening the container and preparing spray wear:

- cotton overalls buttoned to the neck and wrist and a washable hat,
- elbow-length PVC gloves and
- goggles

When using (except aerial spraying) the prepared spray, wear

- cotton overalls buttoned to the neck and wrist and a washable hat and
- elbow-length PVC gloves. Wash hands after use.

After each day's use, wash gloves, goggles and contaminated clothing.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance:	White to off-white granules
Odor:	With moldy odor
Melting Point:	203 °C °C (<i>a.i.</i>)
Boiling Point:	Decompose before boiling (<i>a.i.</i>)

Bulk Density:	0.68 - 0.78 g/cm ³ (approx.)
pH:	5.0 – 8.0
Solubility in water:	Disperse in water
Vapor Pressure:	2×10 ⁻³ mPa (25 °C) (<i>a.i.</i>)

SECTION 10. STABILITY AND REACTIVITY

Stability:	Stable under normal use and storage.
Hazardous Polymerization:	Will not occur.
Conditions to Avoid:	None known.
Materials to Avoid:	Alkaline materials.
Hazardous Decomposition Products:	Combustion or thermal decomposition will evolve toxic and irritant vapor.

SECTION 11. TOXICOLOGICAL INFORMATION

Acute Toxicity Data for the Formulation:

Acute Oral Toxicity	Rat: 50 mg/kg < LD50 < 300 mg/kg.
Acute Dermal Toxicity	Rat: LD50: > 2000 mg/kg.
Acute Inhalation Toxicity	Rat: LC50 (4 h): 0.26 mg/L air.
Eye Irritation	Rabbit: Non-irritating to eyes.
Skin Irritation	Rabbit: Non-irritating to skin.
Sensitization	Guinea pig: Not a skin sensitizer.

SECTION 12. ECOLOGICAL INFORMATION

Ecological Information (for Fipronil Technical)

Bird: Acute oral LD50 for bobwhite quail 11.3, mallard ducks > 2000, pheasants 31, red-legged partridges 34, house sparrows 1120, pigeons > 2000 mg/kg. Dietary LC 50 (5 d) for bobwhite quail 49, mallard ducks > 5000 mg/kg diet.

Fish: Acute LC50 (96 h) for bluegill sunfish 85, rainbow trout 248, European carp 430 µg/L.

Daphnia: LC50 (48 h) 0.19 mg/l.

Algae: EC50 (96 h) for *Scenedesmus subspicatus* 0.068 mg/l; (120 h) for *Selenastrum capricornutum* >0.16, *Anabaena flos-aquae* >0.17 mg/l.

Bees: Highly toxic to honeybees both by direct contact and by ingestion. However no risk to bees when used as a soil or seed treatment.

Worms: LC50 for *Eisenia fetida* > 1000 mg/kg dry soil.

Soil/Environment (for Fipronil Technical)

Readily degraded; major degradates in soil (aerobic) are sulfone and amide, (anaerobic) are sulfide and amide. Photolysis of soil-applied fipronil gives the photodegradate together with sulfone and amide. Both fresh and aged column leaching studies (5 soils) indicate that fipronil and its metabolites present a low risk of downward movement in soil; this is supported by field dissipation studies.

SECTION 13. DISPOSAL CONSIDERATIONS

Single rinse before disposal. Add rinsings to spray tank. DO NOT dispose of undiluted chemicals on site. Puncture or shred and bury empty containers in a local authority landfill. If no landfill is available, bury the containers below 500 mm in a disposal pit specifically marked and set up for this purpose clear of waterways, desirable vegetation and tree roots. Empty containers and product should not be burnt.

SECTION 14. TRANSPORT INFORMATION

Land transport ADR/RID

UN number:	2588
Proper Shipping Name:	Pesticide, solid, toxic, n.o.s. (contains fipronil)
Class:	6.1
Subsidiary Risk:	Not regulated
Packaging group:	III

Marine transport IMDG

UN number:	2588
Proper Shipping Name:	Pesticide, solid, toxic, n.o.s. (contains fipronil)
Class:	6.1

Subsidiary Risk: Not regulated

Packaging group: III

Marine pollutant: Yes

Air transport ICAO/IATA

UN number: 2588

Proper Shipping Name: Pesticide, solid, toxic, n.o.s. (contains fipronil)

Class: 6.1

Subsidiary Risk: Not regulated

Packaging group: III

SECTION 15. REGULATORY INFORMATION

EPCRA SARA Title III Classification:

Section 311/312 Hazard Classes: Acute Health Hazard
Chronic Health Hazard

Section 313 Toxic Chemicals: Not applicable

California Proposition 65

Not applicable

CERCLA/SARA 302 Report Quantity (RQ)

None

RCRA Hazardous Waste Classification (40 CFR 261)

Not applicable

TSCA Status

Exempt from TSCA, subject to FIFRA

SECTION 16. OTHER INFORMATION

MSDS Number: SN581/80%WG

Revise date: January 18, 2020

Revision Summary:

To reflect database modification (no substantive changes - only related to system).

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